ModelIt!

Student Activity Pack

The Quantum Computing Revolution

Modeling Qubit Behavior, Superposition, and Computational Advantage

Grade 11L3 | NGSS HS-PS4-3, HS-PS4-5 | Scale, Proportion, and Quantity

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What is the difference between a classical bit (0 or 1) and a quantum bit (qubit), and why does that difference matter for computing?

2 Why do you think quantum computers need to be cooled to nearly absolute zero to function?

3 Draw a diagram showing how you think a quantum computer processes information differently from a regular computer.

4 What types of problems do you think quantum computers could solve that regular computers cannot?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Qubit Count | Coherence Time | Gate Error Rate | Entanglement Fidelity | Operating Temperature | Circuit Depth | Quantum Volume

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Current State-of-the-Art

1,000 qubits, 100 microsecond coherence, 0.3% gate error rate

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Error-Corrected Regime

10,000 qubits, 0.01% gate error, 10 ms coherence time

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Thermal Disruption

Operating temperature increases from 15 mK to 100 mK

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Qubit Connectivity — The degree to which qubits can directly interact with each other on the processor chip — limited connectivity forces extra gate operations to move information between distant qubits, consuming precious coherence time

Readout Fidelity — The accuracy with which the final quantum state of each qubit can be measured — measurement errors in the final readout can negate the computational advantage built up during the quantum circuit

Cross-Talk — Unwanted quantum interactions between neighboring qubits caused by their physical proximity on the chip — cross-talk creates correlated errors that are harder to correct than independent random noise

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why would a computer built on the strangest rules in physics solve problems that every supercomputer on Earth combined could never crack? Use evidence from your simulation to support your answers.

1 How does your model demonstrate the exponential scaling of quantum computational power versus the exponential scaling of errors?

2 Why is the error correction overhead such a critical barrier to practical quantum computing?

3 What does your model reveal about the relationship between operating temperature and every other quantum parameter?

4 How would your model change if a room-temperature quantum computing technology were developed?

5 What are the limitations of modeling quantum advantage when the problems quantum computers solve best are the ones classical computers cannot verify?

8 STEM Challenge: Design a Quantum Computing Application Proposal

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YOUR ENGINEERING MISSION

Identify a real-world problem where quantum computing could provide transformative advantage over classical methods and design a proposal specifying the quantum resources needed, the expected speedup, and the timeline for feasibility.

THE SCENARIO

A technology investment firm wants to identify the most promising near-term applications for quantum computing. Your team must select a specific problem domain — drug discovery, materials science, logistics optimization, cryptography, or climate modeling — and design a proposal that quantifies the quantum resources needed, estimates when hardware will be sufficient, and calculates the economic value of solving this problem faster than classical computers can.

Engineering Design Thinking

1. WHAT QUANTUM ALGORITHM WOULD YOUR APPLICATION USE AND WHY IS IT FASTER THAN THE BEST CLASSICAL ALTERNATIVE?
2. WHAT ARE THE MINIMUM QUBIT COUNT, COHERENCE TIME, AND GATE FIDELITY REQUIREMENTS FOR YOUR APPLICATION?
3. HOW WOULD YOU VERIFY THAT THE QUANTUM COMPUTER PRODUCED A CORRECT ANSWER FOR A PROBLEM TOO HARD FOR CLASSICAL VERIFICATION?
4. WHAT IS THE ESTIMATED ECONOMIC VALUE OF SOLVING THIS PROBLEM WITH QUANTUM ADVANTAGE VERSUS WAITING FOR CLASSICAL HARDWARE IMPROVEMENTS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Quantum Software Engineers design algorithms and applications for quantum computers at companies like IBM Quantum, Google Quantum AI, IonQ, and Rigetti Computing, earning $120,000-$200,000/year. Quantum Hardware Engineers who build and maintain quantum processors work in cryogenics, microwave engineering, and materials science, earning $100,000-$180,000/year. Quantum Information Scientists at research universities and national labs push the theoretical boundaries of what quantum computers can achieve, earning $90,000-$170,000/year.